ANSWER KEY & MARKING SCHEME SET A

Chapter 1: **Exploration: Entering the World of Chemistry** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** a simplified way of looking at a real system that keeps only what matters for the question	models ignore details on purpose
2	**(a)** spheres joined by bonds	a model for structure
3	**(d)** pounds and kilograms	unit mix-up in density
4	**(b)** a law describing regular behaviour	pattern in motion
5	**(c)** re-examine assumptions, models or measurements	mismatch drives further exploration
6	**(d)** check whether an answer is reasonable	estimation builds intuition and detects errors
7	**(c)** has a specific meaning different from everyday use	scientific terms are precise
8	**(a)** physics, chemistry and biology together	interdisciplinary problem
9	**(b)** is backed by careful testing and evidence	evidence-based
10	**(d)** ignore individual cells and treat the heart as a working system	keep what matters

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A simplified way of looking at a real system that focuses only on what is important for a question (1). Example: a moving car treated as a single point (1).	1+1
12	The choice is made on purpose to keep the problem simple enough while still answering the question (1); extra details can be added later for accuracy (1).	1+1
13	Matter: mass of ball, speed and direction of the hit. Can be ignored: brand of bat, colour of ball, grass on the field (1+1).	1+1
14	So that scientists everywhere can describe observations, compare results and build ideas together without confusion (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Real systems are too complex to study in full detail (1). A model keeps only the key factors, e.g. falling object: ignore air resistance to study gravity (1). Assumptions simplify; they are added back for accuracy later (1).	1+1+1
16	Important: mass of ball, speed, direction (1). Minor effects: air resistance, spin, seam (1). Irrelevant: bat brand, ball colour, grass (1).	1+1+1
17	Keep: distance, average speed, stops at crossings (1+1). Ignore: colour of cycle, school name, tyre brand (1)—they do not change the time.	2+1
	OR: Everyday use is loose (effort, tiring activity) (1); in science each term has a precise definition with symbols and units so that results can be compared (2).	1+2
18	c comes from Latin celeritas = speed (1); symbols come from history and international agreement (1); this lets scientists across the world use the same language (1).	1+1+1
	OR: Ground crew used density in pounds per litre instead of kilograms per litre (1); fuel was about 15 000 litres short (1); using SI units everywhere avoids conversion errors (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) A unit conversion error (1). (ii) It had about 15 000 litres less fuel than required (1). (iii) Standard units everywhere avoid conversions and mistakes (2). (iv) Mass and volume (or density and fuel quantity) (1).	1+1+2+1
	OR: (i) Mass of ball, speed and direction (1). (ii) Bat brand, colour, grass (1). (iii) Air resistance, spin, seam of the ball (1). (iv) A simplified representation keeping key features and ignoring minor ones (2).	1+1+1+2

20	(i) Law: describes a pattern; theory: explains why; principle: broad idea (2). (ii) Newton's laws; atomic theory; conservation of energy (2). (iii) A theory is based on careful testing and evidence and can be improved (1).	2+2+1
	OR: (i) It asks for measurable evidence and past patterns (1). (ii) Humidity at last rain; wind speed/direction; temperature drop (1+1). (iii) Tiny changes in temperature, pressure, humidity and wind grow with time (1). (iv) Re-examine assumptions, models or measurements (1).	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET B

Chapter 1: **Exploration: Entering the World of Chemistry** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** a deliberate simplification to understand the basic effect of gravity	simplifying assumptions are part of modelling
2	**(c)** a hot gas, focusing on temperature and pressure	he ignored many complex processes
3	**(b)** results can be compared and conversion errors are avoided	agreed international standards
4	**(b)** how molecules are formed	an explanation based on evidence
5	**(b)** What was the humidity the last time it rained?	measurable evidence and past patterns
6	**(b)** 20 thousand	about 14 × 1440 ≈ 20 000
7	**(a)** a language to express and test relationships clearly	equations are compact statements
8	**(c)** careful observation	as explained in the chapter
9	**(d)** Rejecting an idea on the basis of opinion alone	ideas are rejected on evidence
10	**(d)** Predictions that fail lead to better ideas	self-correcting nature

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The fuel was miscalculated; the aircraft was about 15 000 litres short and had to glide to an emergency landing (2).	2
12	A law describes a regular pattern in nature (1); a theory explains why the pattern occurs, based on evidence (1).	1+1
13	In science a theory is an explanation based on careful testing and critical examination (1). It may change with new evidence but it is not an untested guess (1).	1+1
14	A reasoned expectation about what will happen based on laws, theories, models and evidence (1); a guess has no such basis (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Law: describes a pattern, e.g. Newton's laws (1). Theory: explains why, e.g. atomic theory (1). Principle: broad idea, e.g. conservation of energy (1).	1+1+1
16	Theories are tested against new evidence (1); when predictions fail, scientists change assumptions or models (1); this openness to correction lets science improve (1).	1+1+1
17	Questions on humidity (above 80% last time?), wind speed and direction, temperature trend (1 each) — they use data and past patterns, not just ‘dark clouds’.	1+1+1
	OR: Ask what happens during an eclipse (the Moon blocks sunlight) (1); does temperature or any chemical/biological change in food occur? measure it (1); no mechanism or evidence is found, so the claim is unsupported (1).	1+1+1
18	Weather depends on temperature, pressure, humidity, wind (1); small differences grow with time, so long-range forecasts become uncertain (2).	1+2
	OR: Breaths per minute ≈ 14 (1); breaths per day ≈ 20 000 (1); volume per breath ≈ 0.5 litre ⇒ about 10 000 litres per day (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Estimate breaths per minute, per day and volume per breath (1). (ii) 14 × 1440 ≈ 20 000 breaths; 0.5 L per breath ⇒ about 10 000 L (2). (iii) 3 balloons × 2 L = 6 L/min ⇒ 8640 L per day; close to 10 000 L so the estimate is reasonable (2).	1+2+2

	OR: (i) The Moon blocks sunlight reaching the Earth (1). (ii) Temperature, spoilage, chemical or biological changes in the food (1). (iii) There is no mechanism or evidence, so the claim is not supported (2). (iv) It prevents false beliefs; ideas are accepted on evidence (1).	1+1+2+1
20	(i) Making a rough, reasoned calculation to check whether a result is reasonable (1). (ii) 2000–2500 kcal per adult per day; use kcal per 100 g of rice to find grams per day and per month (2). (iii) 100 g is far too little; a few tonnes far too much; an answer of tens of kilograms is sensible (2).	1+2+2
	OR: (i) Pressure raises the boiling point of water (1). (ii) Cooking changes food chemically; nutrients/microbes are affected (1). (iii) Relationships between pressure, temperature and time can be expressed and tested using equations (2). (iv) Real problems cross boundaries made by humans to organise knowledge (1).	1+1+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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